

# Silver Full Build Notebook

This notebook is for loading initial datasets, should not need to be reran.

## Loading Data:

Reading in initial data. This data is already loaded into the OneLake -> NCPSS\_registry -> Files -> bronze

If new initial data needs to be reloaded into the lakehouse, it should be added to the bronze file to be called upon from within a notebook.

From the Bronze folder we are reading in the Drupal csv and Access Excel and turning them into dataframes to be used throughout the notebook.

Some initial data analysis was done to ensure that the read in was successful and to see what the column data types are.

Columns are renamed to make standardizing and eventually merging the data easier.

## Data Cleaning Section:

Cleaning Date of Birth column, adjusting column types, editing string columns where typos persisted. Zip codes standardized to only 5 digits, null values made null data type, values made lowercase and trailing spaces trimmed.

Wage columns cleaned to all be type double, this accounts for decimal places. Column specific errors are dealt with here: ie) gender, ethnicity, education and county data.

Unique identifier made for columns to be used later.

## Data Merge:

A unique fingerprint for each row was generated using hash to ensure uniqueness and check for duplicates across rows.

Merging was done between the two datasets, merging first by email, next by first name, last name, and DOB, and thirdly by first name, last name, and zip code. Removes duplicates and then adds in unmatched rows from both Drupal data set and Access data set. Prints into one final merged dataset.

A source record column is added for documentation throughout use.

## Data Review:

After merge various code blocks are run to ensure that the merge was successful and data combined is sensible and without error.

## Saving Cleaned Data Into Table in Fabric:

Data is exported into a table within the data lake tables -> dbo. This table can be called upon and used for the gold notebooks.